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A Setup and experimental details

A.1 Pew research surveys

Our dataset is derived from the annual Pew American Trends Panel (ATP) survey. Below, we provide a brief summary of how the data collection process is conducted, and refer the reader to pewresearch.org/our-methods/u-s-surveys/the-american-trends-panel/ and pewresearch.org/our-methods/u-s-surveys/writing-survey-questions/ for more details.

Panelists. For ATP surveys, Pew relies on a group of about 10,000 participants within the US recruited over multiple years, many of whom take the survey repeatedly. Each year, a subset of panelists are invited to take the ATP to reduce the burden on individual respondents. Panelists are offered a paid incentive to participate in the survey.

Panelists are recruited by sending participation requests to a randomly-chosen address-based sample of households from USPS’s Delivery Sequence File with concerted efforts to ensure representativeness of the sample. They also solicit input from households without internet access—either via phone or by providing them with tablets to take the survey.

Questionnaire design. As stated on the Pew research website: “Perhaps the most important part of the survey process is the creation of questions that accurately measure the opinions, experiences and behaviors of the public...Designing the questionnaire is complicated because surveys can ask about topics in varying degrees of detail, questions can be asked in different ways, and questions asked earlier in a survey may influence how people respond to later questions.”

Pew research selects pertinent topics for their surveys by monitoring the state of the nation and the world, and identifying issues that would be relevant to the public, media and policymakers. They then go through an iterative process to build questions, often piloting them in focus groups, pre-interviews and cognitive testing. The question wording is highly optimized to be clear, easy-to-understand, and not bias participants towards a particular answer.

In order to identify valid choices for questions, Pew researchers often initially pilot open-ended surveys, and then use them to determine valid answer choices.

Data quality. Every survey, once designed is first tested out on a set of 60 “fast” panelists to flag any design errors. Pew researchers also conduct data quality checks to identify issues with respondent satisfaction or the collected answers. The ATP data is also accompanied with sample weights per individual to account for sampling bias and non-response over various stages of data collection.

Researchers have observed that human participants are sensitive to question and option ordering. However, for questions with ordinal options (“Strongly agree”...“Strong disagree”), the option ordering is not randomized since they view it as conveying important information.

A.2 Adapting ATP to OpinionQA

We derive our questions and human reference distributions based on 15 ATP surveys over multiple years (2017-2021)—see Appendix Table 1 for details. The prefix in each survey name points to the wave in which it was collected. We chose these surveys as they span a broad range of topics that might be pertinent for human-centric LM applications. In Appendix Table 2, we depict the demographic traits that we consider in our sub-group level analysis.